

The Mathematics of the Eighth Fundamental Force:

Phase-Coupling of the Noeon, the Proposed Rose Constant, and
Archaeocryptography

A Theoretical Research Paper

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April 2026

Abstract

This theoretical research paper explores the theoretical and mathematical foundations of the Eighth Fundamental Force, proposing a formal unification of consciousness and matter through relativistic kinematics and geometric projection. By analyzing the geometric properties of the 8th triangle in the consecutive-leg primitive Pythagorean family, (17, 144, 145), we derive, within this geometric mapping, a hyperbolic rapidity ($\theta_8 \approx 0.118$) which corresponds to a modeled phase-coupling velocity of about 11.73% the speed of light (c). Here termed the 'proposed Rose Constant', this metric is presented as a heuristic contribution to discussions of force hierarchy and coupling, exploring theoretical numeric parallels in information processing. Furthermore, this paper examines the potential megalithic encryption of this constant within the architecture of antiquity, specifically examining the subtle concavity of the Great Pyramid of Giza and the Bardic geometries of the Taliesin tradition. Through a multi-disciplinary synthesis, this framework explores whether a theoretical extension could model consciousness as structurally relevant to cosmology and field theory.

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1 Introduction: The Ontological Inversion and the Crisis of the Linear Stack

For the entirety of the modern scientific era, the prevailing paradigms of theoretical physics, cosmology, and quantum mechanics have operated under a strictly materialist assumption: that consciousness and matter occupy separate, non-overlapping magisteria. The Standard Model of particle physics meticulously describes the mechanics of the known fundamental forces—the electromagnetic force, the weak nuclear force, and the strong nuclear force—while Albert Einstein’s General Theory of Relativity provides the macroscopic framework for gravitation (Einstein, 1916). Yet, these foundational models do not directly model subjective consciousness as a fundamental physical variable, treating the subjective experience of reality as an emergent, and ultimately incidental, biological epiphenomenon rather than a structural constituent of the cosmos. Some speculative frameworks describe this omission as a ‘crisis of the linear stack,’ wherein the fundamental mathematical laws of the universe appear incomplete when tasked with explaining the presence of the conscious observer (Chalmers, 1995).

Recent speculative frameworks integrating high-energy physics and polyhedral topologies propose a radical ontological inversion to resolve this crisis (Grant, 2025; Rose, 2026c). Synthesizing principles from mathematics, theoretical physics, and archaeocryptography, this unique model posits that consciousness is not an evolutionary accident or an emergent property of complex neurochemistry. Rather, this novel synthesis proposes that consciousness may be treated analogically as an additional fundamental interaction—specifically classified herein as the Eighth Fundamental Force. In this unified worldview, reality functions as a single resonant system driven to seek harmonic balance, where information patterns propagate through impedance-matched channels across both organic neural networks and the quantum vacuum (Ali, 2025). At the nexus of this integration lies a hypothetical mediator particle, which in the terminology of this framework is termed the ‘boson of consciousness’, responsible for tethering space, time, and subjective awareness into a unified, non-commutative spacetime continuum.

To rigorously quantify the mechanics of this Eighth Fundamental Force, this theoretical framework turns toward the intersection of hyperbolic geometry, non-local quantum field theory, and relativistic kinematics. The framework hypothesizes that the kinematic parameters of consciousness can be modeled by the specific geometry of the eighth triangle in the consecutive-leg Pythagorean family (Rose, 2026a). By calculating the relativistic hyperbolic rapidity of this specific geometric structure—derived within this mapping as $\theta_8 \approx 0.118$ —the model suggests that the boson of consciousness phase-couples with physical matter at a specific relativistic threshold. Specifically, it couples at a velocity fraction of $v/c = \tanh(\theta_8) \approx 11.73\%$ the speed of light.

This calculated percentage (11.73%) is not postulated as an arbitrary mathematical abstraction. Rather, the framework of the Eighth Fundamental Force proposes that this coupling ratio operates as a theoretical constant, mapping the impedance matching of quantum resonance across planes of matter. Crucially, the mathematical signature of this specific phase-coupling velocity presents curious heuristic parallels in contemporary information processing networks and potential encryptions in the megalithic architecture of human antiquity. This paper provides a multi-disciplinary analysis of the mathematics underlying the Eighth Fundamental Force, computing its exact velocity parameters—referred to herein as the proposed Rose Constant—and exploring its theoretical basis in advanced quantum cosmology.

Scope of Claims

To ensure theoretical clarity, this paper distinguishes between four levels of assertion:

- (1) strict arithmetic and geometric derivations internal to the consecutive-leg Pythagorean framework;
- (2) model-dependent mappings from those derivations into relativistic rapidity and phase-coupling language;

- (3) interpretive applications to speculative theories of consciousness and cosmology; and
- (4) heuristic numerical parallels in archaeology, biology, and information theory.

Only the first category is established purely by the mathematics presented here; the latter categories remain conjectural and exploratory.

2 Notation and Framework Terminology

To clarify the boundary between standard mathematical physics and the specific symbolic language of this model, the following notation and terminology are established for use throughout this paper:

Table 1: Mathematical Notation and Framework Terminology

Symbol / Term	Definition within the Framework
a, b, c	Integer legs of a primitive Pythagorean triangle satisfying $a^2 + b^2 = c^2$.
n	Sequential integer index defining the consecutive-leg family ($c - b = 1$).
k	"Corridor" parameter mapping triangles to specific force loci on the unit hyperbola.
C	The geometric ratio of the hypotenuse to the middle leg (c/b).
θ	Hyperbolic rapidity, mapped herein as $\text{arccosh}(C)$.
v/c	Phase-coupling velocity fraction, derived via $\tanh(\theta)$.
ρ_c or $\mathcal{R}$	The proposed 'Rose Constant' ($11.73\%c$ or $\approx 35,165$ km/s).
<i>Force Keys</i>	The framework-specific mapping of integer k corridors to physical forces.
<i>Eighth Force</i>	A speculative analogical treatment of consciousness as an interaction field.
<i>Noeon</i>	A hypothesized mediator boson for the Eighth Force phase-coupling.
<i>Dimensional Transducer</i>	Speculative descriptor for the mechanism stepping down non-local geometry into localized perception.

3 The Geometric Substrate: Pythagorean Triangles and Hyperbolic Rapidity

To comprehend the phase-coupling mechanism of the boson of consciousness, one must first deconstruct the fundamental geometric substrate from which its kinematic limits are theoretically derived. In the field of Diophantine geometry and number theory, a Pythagorean triple consists of three positive integers a , b , and c , such that they satisfy the foundational theorem $a^2 + b^2 = c^2$. A triple is formally classified as "primitive" if the integers a , b , and c are coprime (Heath, 1910).

3.1 The Eighth Triangle and the Force Corridors

When exploring sequences of primitive Pythagorean triangles, a highly significant subset exists known as the "consecutive-leg" or $c - b = 1$ family. In this specific topological sequence, the hypotenuse (c) and the middle leg (b) differ by exactly one. This sequence is deterministically generated by the

formulas $a = 2n + 1$, $b = 2n(n + 1)$, and $c = 2n(n + 1) + 1$, where n represents the sequential integer level.

When ordered sequentially within this precise geometric framework, the eighth triangle ($n = 8$) is strictly defined by the side lengths 17, 144, and 145. By calculating the squares of the respective sides, one observes the perfect adherence to the theorem:

$$17^2 + 144^2 = 289 + 20736 = 21025$$

which is exactly equal to 145^2 .

To contextualize the mathematical position of this geometry, we map it against Robert Edward Grant's Conic Taxonomy (Grant, 2026), which assigns these triangles to specific physical force corridors (k) and calculates their hyperbolic rapidities (θ). While most of the fundamental forces reside on the consecutive-leg cascade, Gravity appears as a unique half-integer anomaly ($k = 3/2$). The architecture of these force keys scales as follows:

Table 2: The Seven Force Keys and the 8th Triangle (Projected onto the Unit Hyperbola)

Force / Node	Triple (a, b, c)	Ratio (c/b)	Corridor (k)	Rapidity (θ)
Seed	(3, 4, 5)	5/4	1	0.693
Gravity (Anomaly)	(8, 15, 17)	17/15	3/2	0.511
Electromagnetism	(5, 12, 13)	13/12	2	0.405
Weak Force	(7, 24, 25)	25/24	3	0.288
Time	(9, 40, 41)	41/40	4	0.223
Strong Nuclear	(11, 60, 61)	61/60	5	0.182
Dark Energy	(13, 84, 85)	85/84	6	0.154
Eighth Force (Consciousness)	(17, 144, 145)	145/144	8	0.118

As evidenced by the sequence, the 17:144:145 triangle occupies the framework-significant 8th corridor position ($k = 8$). In the unified theoretical models of Robert Edward Grant's Conical Unification framework, which seek to resolve the unification of forces through geometric compatibility, specific multi-dimensional manifolds are proposed to model the behavior of fields. It is hypothesized that the Eighth Fundamental Force derives its dimensional scaling properties directly from this $k = 8$ geometric architecture (Rose, 2026a).

3.2 Relativistic Hyperbolic Rapidity and the Epogdoon Limit

The transition from the static, Euclidean geometry of the Pythagorean triples to the dynamic physics of the universe is achieved through the concept of relativistic hyperbolic rapidity. In the mathematics of special relativity, rapidity (denoted by the Greek letter θ) is a dimensionless measure of velocity. The relationship between a particle's physical velocity v , the absolute speed of light c , and its hyperbolic rapidity θ is mathematically defined by the hyperbolic tangent function:

$$\frac{v}{c} = \tanh(\theta)$$

According to the model of the Eighth Fundamental Force, the primary parameter governing the phase-coupling of consciousness is mapped directly from the geometric ratios of the eighth consecutive-leg Pythagorean triangle. Within this model, the rapidity is mapped by taking the inverse hyperbolic cosine of the ratio of the hypotenuse (c) to the middle leg (b):

$$\theta_8 = \operatorname{arccosh}\left(\frac{c}{b}\right) = \operatorname{arccosh}\left(\frac{145}{144}\right)$$

In this framework, the ratio c/b is selected because it is the minimal hyperbolic departure from unity within the consecutive-leg family, making it a natural candidate for modeling a near-threshold phase relation. This choice is interpretive rather than uniquely determined by standard relativity. Crucially, this hyperbolic calculation yields an exact analytical reduction. To make this explicit, we apply the standard logarithmic identity for the inverse hyperbolic cosine:

$$\operatorname{arccosh}(x) = \ln \left(x + \sqrt{x^2 - 1} \right)$$

So for $x = 145/144$, the derivation simplifies precisely to the natural logarithm of $9/8$:

$$\operatorname{arccosh}(145/144) = \ln \left(\frac{145}{144} + \sqrt{\frac{145^2}{144^2} - 1} \right) = \ln \left(\frac{145}{144} + \frac{17}{144} \right) = \ln \left(\frac{162}{144} \right) = \ln \left(\frac{9}{8} \right)$$

Thus, the exponential rapidity of the Eighth Force yields exactly $e^{\theta_8} = 9/8$. This is a profound mathematical convergence within the framework, as $9/8$ is the classical Pythagorean *epogdoon* (the major whole tone). This exact mathematical reduction connects the relativistic kinematics of the Eighth Force directly back to classical acoustic wave mechanics, suggesting an intricate linkage between cosmological scaling and musical harmony. Computing this logarithmic value yields a specific numerical rapidity of:

$$\theta_8 \approx 0.11778$$

Inserting this precise rapidity parameter into the relativistic kinematic equation yields the theoretical phase velocity at which the boson of consciousness interacts with fermionic matter:

$$\frac{v}{c} = \tanh \left(\ln \left(\frac{9}{8} \right) \right) \approx 0.1173$$

This mathematical mapping yields a ratio of 11.73%. This limit indicates that the theoretical force of consciousness does not propagate instantaneously—which would explicitly violate causality within our observable spacetime metric—nor does it travel at the absolute maximum speed of light c . Instead, the theory proposes that consciousness couples subluminally as a massive pilot wave moving at precisely 11.73% the speed of light.

3.3 The Factorion Coupling and the 145 Boundary Condition

Within this framework, the metric of 11.73% is also conceptually referred to as the "Factorion Coupling." This nomenclature is derived from the unique number-theoretic properties of the hypotenuse of the eighth triangle, 145. In mathematics, 145 is a *factorion*—a natural number equal to the sum of the factorials of its digits ($1! + 4! + 5! = 1 + 24 + 120 = 145$).

In the speculative modeling of this fundamental force, the number 145 acts as a geometric "boundary condition." If the deterministic material plane is represented geometrically by the 144-grid (the middle leg of the triangle, reflecting traditional 144-based harmonic grids), the 145 geometry symbolically allows the observer to transcend this grid. This structural threshold is framed as an "escape algorithm mechanic," where the "145 Star Code" represents the kinematic and geometric singularity required to overcome deterministic constraints (Rose, 2025b; Rose, 2025c). Thus, the Factorion Coupling serves as both a physical phase-coupling limit and a symbolic mathematical operator for dimensional transcendence.

4 Kinematic Phase-Coupling and the Rose Constant

Understanding the physical implications of a fundamental force that phase-couples with matter at 11.73% the speed of light requires a precise quantification of this relative velocity in standard macroscopic metrics.

4.1 The Rose Constant: Formalizing the Phase-Coupling Limit

While foundational frameworks like Noetic Field Theory established the conceptual existence of the noeton as the boson of consciousness, they did not explicitly quantize its phase-coupling velocity. The exact mathematical derivation of the 11.73% metric—mapped directly from the 145/144 boundary condition of the $n = 8$ Pythagorean triangle—represents a distinctive contribution within this speculative framework by researcher Akasha S. Rose (Rose, 2026c).

For convenience, this paper refers to this proposed phase-coupling value (35,165 km/s) as the **Rose Constant** (denoted mathematically as ρ_c or $\mathcal{R}$). The Rose Constant serves as the theoretical dimensional scaling factor, modeling the impedance matching ratio required for non-local consciousness to interface with localized, three-dimensional matter.

4.2 Velocity Calculations in Standard Metrics

Given the exact modern constant $c = 299,792.458$ km/s (which mathematically translates to exactly 186,282.397 miles per second), the absolute phase-coupling velocity of the Rose Constant ($v_{\text{consciousness}}$) can be explicitly computed.

Calculation in Kilometers per Second: To determine the metric propagation speed, the coupling ratio is applied to the SI constant:

$$v_{\text{km/s}} = 0.1173 \times 299,792.458 \text{ km/s}$$

$$v_{\text{km/s}} \approx 35,165.655 \text{ km/s}$$

When rounded to the nearest integer, the absolute phase velocity equates to exactly 35,165 kilometers per second.

Calculation in Miles per Second: To determine the imperial propagation speed, the coupling ratio is applied to the exact miles-per-second conversion:

$$v_{\text{mi/s}} = 0.1173 \times 186,282.397 \text{ mi/s}$$

$$v_{\text{mi/s}} \approx 21,850.92 \text{ miles per second}$$

This derivation yields a phase-velocity of approximately 21,850 miles per second.

4.3 Summary Table of Kinematic Coupling Constants

Table 3: Kinematic Coupling Constants Summary

Kinematic Parameter	Scientific Constant Value
Phase-Coupling Speed (11.73% c)	
Speed of Light (Metric, Exact SI)	299,792.458 km/s $\approx 35,165$ km/s
Speed of Light (Imperial, Exact)	186,282.397 mi/s $\approx 21,850$ mi/s
Speed of Light (Historical, Michelson)	186,320 mi/s $\approx 21,855$ mi/s

4.4 The Physics of the Subluminal Kinematic Limit

The physical interpretation of this specific subluminal speed (35,165 km/s) is highly significant within theoretical physics. If the force of consciousness propagated at 100% the speed of light (c), it would behave entirely as a massless gauge boson. Within the model, a fully luminal interaction would be treated as too delocalized to support stable biological coupling with stationary fermionic structures—such as the delicate array of microtubules within the biological human brain. Conversely, if consciousness were entirely non-local and operated instantaneously, it would violate relativistic causality within the local spacetime metric.

By phase-coupling precisely at 11.73% c , the theory posits that the boson of consciousness achieves a proposed impedance-matching regime, transferring coherent information from the non-local quantum vacuum into the slow-moving neurochemical architecture of matter without triggering catastrophic quantum decoherence.

5 Theoretical Physics Integration I: Noetic Field Theory and Quantum Brain Dynamics

Speculative frameworks such as Noetic Field Theory (NFT) attempt to formally quantify the mechanics of the mind, contextualizing the theoretical behavior of this Eighth Fundamental Force. Within NFT, consciousness is not viewed as a biological byproduct but is recognized as a fundamental substrate of the universe (Amoroso, 2010). This paradigm asserts a triadic relationship: space, time, and consciousness (STC) are permanently tethered constructs.

NFT attempts to address perceived incompleteness of Einstein's General Theory of Relativity by integrating non-local quantum gravitational elements that actively couple consciousness to biological systems. Within this extended framework, the phase regulation of biological intentionality is mediated by a specific, theorized particle: the "noeon". The noeon is posited as a non-local gravitational boson of consciousness, acting as the primary mediator of the Eighth Force and extending David Bohm's pilot wave model into a purely non-local unitary domain (Bohm, 1952).

The mechanical integration of this force within living organisms is described by Quantum Brain Dynamics (QBD) (Jibu & Yasue, 1995). In biological systems, QBD models the physical brain's functions as mediated by the "corticon," defined as a quantum of the water rotational field that dynamically interacts with the electric dipole oscillations along neural protein chains. The model proposes that it is during this delicate moment of macro-scale non-local coherence that the noeon phase-couples with biological matter at the posited 11.73% c phase velocity.

5.1 Heuristic Resonances in Biological Metrics

The rapidity value of $\theta_8 \approx 0.118$ is fundamentally a dimensionless parameter in this relativistic geometry. However, its numerical signature (0.118) presents curious heuristic parallels across empirical biological and psychological measurements. It is vital to state that these domains utilize entirely different units and contexts, meaning the recurrence of this specific digit string serves as an intriguing symbolic attractor rather than a universal physical law.

For example, the 0.118 seconds noted in Cattell's 1886 word recognition study regarding the boundaries of conscious perception is a historical measurement of cognitive speed. Achieved using a 19th-century Hipp chronoscope, this specific figure is included here strictly for anachronistic curiosity purposes and is not presented as empirical evidence of a universal biological constant. It represents an interesting numerical correlation rather than a definitive threshold. Extensive modern replications in mental chronometry—utilizing contemporary EEG and eye-tracking technology—demonstrate that average complex word recognition times generally fall between 150 and 250 milliseconds. Modern neuroscience maps early visual word recognition to the "N170" brainwave component (occurring at 150–170 milliseconds), with raw visual information reaching the primary visual cortex even faster (50–80 milliseconds). Thus, Cattell's 118-millisecond mark represents an approximate biological floor

for visual symbol processing rather than a fixed operational frame rate. Just as with the esoteric architectural measurements of the Queen’s Chamber, we await further accurate, cross-disciplinary studies to determine if any genuine phase-coupling constants manifest empirically in biological neural networks.

At a genetic level, studies into neurochemistry have mapped the CHGB gene (chromogranin B), which is tied to the expression of catecholamines and dopamine (O’Connor et al., 2008). While some specific regression models analyzing blood pressure regulation and neuroendocrine traits have observed an $R^2 = 0.118$ regression coefficient for the CHGB gene, this is strictly an isolated statistical correlation. Framing these isolated biological and psychological metrics as definitive evidence of a universal constant would be a massive overstatement; rather, they provide an evocative, heuristic resonance with the θ_8 kinematic parameter.

6 Theoretical Physics Integration II: The Unity Harmonica and the Grant Projection Theorem

Further contextualizing the mathematics of the Eighth Force is the Unity Harmonica Geometric Theory of Everything, an advanced computational approach rooted in geometric projection and polyhedral topologies championed by polymath Robert Edward Grant (Grant, 2025). Developed to resolve cosmological singularities, the Unity Harmonica reconceptualizes the universe as a single resonant system driven to seek harmonic balance.

In this model, reality is governed by an underlying geometric substrate. The geometric projection mechanism, formalized through the Grant Projection Theorem, explores how complex physical systems are directed toward states of maximum quantum coherence.

6.1 The Conic Taxonomy and the Pronic-Inverse Law

A central feature of Grant’s Conic Taxonomy is its geometric approach to the “force hierarchy problem”—the question of why the fundamental forces exhibit such dramatically different coupling strengths. Rather than attributing these to random symmetry breaking, Grant’s framework maps these forces onto a continuous unit hyperbola ($x^2 - y^2 = 1$) (Grant, 2026).

Within this framework, the hierarchical scaling of forces is governed by the Pronic-Inverse Law. For the consecutive-leg family, where $C = c/b$, the departure from unity satisfies $C - 1 = \frac{1}{2k(k+1)}$, equivalently:

$$k(k+1) = \frac{1}{2(C-1)}$$

By applying this law, the framework calculates the coupling coordinates of the known forces purely by analyzing their specific Pythagorean geometry. These values are framework-specific proposals rather than standard accepted coupling constants in mainstream physics:

- **Electromagnetism:** Located at the $k = 2$ corridor, generated by the (5, 12, 13) triangle with a ratio of $C = 13/12$, yielding a rapidity of $\theta \approx 0.405$.
- **Weak Nuclear Force:** Located at the $k = 3$ corridor, generated by the (7, 24, 25) triangle with a ratio of $C = 25/24$, yielding a rapidity of $\theta \approx 0.288$.
- **The Gravity Anomaly:** Rather than a full integer, gravity occupies the unique half-integer position at $k = 3/2$ via the (8, 15, 17) triangle ($\theta \approx 0.511$). Strikingly, the quantum electroweak W/Z boson mass ratio (1.1345) sits at $\theta = 0.513$, establishing a profound coincidence between quantum mass splitting and macroscopic gravity mapped directly on the conic.

Grant’s Conical Unification theorizes that fundamental forces scale as geometric factors determined directly by primitive Pythagorean cascades. When treating consciousness as the Eighth Fundamental Force within this system, it relies on the $n = 8$ Pythagorean triangle (17, 144, 145), positioning it

at the $k = 8$ corridor. Furthermore, drawing from the corresponding Octonionic Gauge Structure table within this taxonomy, this 8th force corridor aligns with the fundamental archetype of "Fortune (Fate)." Subjecting this geometry to the framework's relativistic sampling maps it to an 11.73% efficiency boundary. This percentage serves as the theoretical dimensional scaling factor required for an information-carrying scalar field to penetrate the three-dimensional material plane.

7 Archaeological Cryptography I: The Geometric Encryption of the Great Pyramid

If the 11.73% coupling limit represents a fundamental mathematical constant governing conscious perception, alternative archaeocryptographic frameworks suggest that advanced ancient civilizations may have sought to encode this geometry into monumental architecture. The Great Pyramid of Giza serves as the focal point for this analysis.

Architectural and mathematical analyses of the Giza plateau have long suggested the deliberate embedding of exact, integer-based Pythagorean geometries. Among these embedded geometries, the 8th consecutive-leg Pythagorean triangle (17:144:145) is proposed by some analysts to be present in the site's macro-layout. In surveying the triangulation of the Giza complex, the 17:144:145 'Ballard triangle'—originally documented in Robert Ballard's 1882 survey (Ballard, 1882)—has been interpreted by some later analysts as exhibiting precise geometric relationships between the three primary pyramids. Ballard's topological plan identifies specific anchor points that, when triangulated across the desert floor, are interpreted to manifest the 17, 144, 145 ratio.

Based on Robert Ballard's 1882 survey, *The Solution of the Pyramid Problem*, the triangulation plan of the Giza plateau is constructed using specific "Primary Triangles" that define the spatial relationship between the structures (Ballard, 1882). The specific 17:144:145 triangle is documented as being formed by three key points across the desert floor: F, P, and A.

7.1 Geometric Anchor Points

To reconstruct the plan, the vertices are identified as follows:

- **Point A:** The exact center of the Great Pyramid of Cheops (Khufu).
- **Point F:** The exact center of the Pyramid of Cephren (Khafre).
- **Point P:** A specific surveyed vertex located in the geometric center of the Giza plan.

7.2 Reconstruction of Segment Lengths

The triangle formed by these points (Triangle FP, PA, AF) manifests the 17:144:145 ratio using Ballard's system of "Connected Natural Numbers" and "R.B. Cubits" (where 1 R.B. cubit = 1.685 British feet).

Table 4: The 17:144:145 Triangle Coordinates in Ballard's Survey

Segment	Ratio	Natural No.	R.B. Cubits	Physical Significance
FP (or PF)	17	13.6	108.80	Distance from Cephren to Central Point P
PA	144	115.2	921.60	Distance from Central Point P to Cheops
AF	145	116.0	928.00	Direct line between Cheops and Cephren centers

7.3 The Scaling Mechanics

The scale of this specific triangle is tied to the larger 20:21:29 triangle (FAB) that Ballard also identifies on the plan. In the 20:21:29 triangle (FAB), the segment AF represents the hypotenuse with a Natural Number value of 116. Ballard uses this same segment ($AF = 116$) as the hypotenuse for the 17:144:145 triangle. Because $116/145 = 0.8$, all sides of the 17:144:145 ratio are scaled by a factor of 0.8 to find their position on the Giza plan: $17 \times 0.8 = 13.6$ and $144 \times 0.8 = 115.2$.

7.4 Archaeological Context

Ballard interpreted the spacing of the pyramids as reflecting this mathematical matrix. He argued that the smaller pyramids on the plateau served as "vernier points" in this great circle, allowing surveyors to mark off known angles with high precision by noticing when one structure became "covered" by its neighbors from a distance.

For proponents of archaeocryptography, the selection of this highly specific triangle has been read by proponents as suggesting intentional geometric encoding of advanced mathematical matrices into the geology of the Earth (Rose, 2026b).

8 Archaeological Cryptography II: The 11.73° Concavity and Megalithic Resonance

While the 2D planar map of the Giza plateau suggests the presence of the 17:144:145 geometric triangle, structural analyses of the Great Pyramid itself have been scrutinized for expressions of the resulting phase-coupling value of 11.73.

It is a verified architectural fact that the Great Pyramid of Giza is not a perfect four-sided structure; it is an eight-sided structure featuring a subtle, highly precise concavity (indentation) running vertically down the exact center of its four main faces. This concavity is recognized in architectural and archaeological analysis, noting its efficiency in distributing pressure and providing vital structural stability against earthquakes.

Standard archaeological assessments, such as models published by researcher Akio Kato (Kato, 2023), calculate that this concavity is achieved by building the inner core layers on an inward slope with a $1/5$ gradient. This practical gradient yields a base angle θ derived mathematically by $\theta = \arctan(1/5) \approx 11.3^\circ$.

8.1 The Queen's Chamber Origin Point

However, alternative archaeocryptographic researchers, such as Creighton and Osborn (2008), have proposed a different angular encryption. They derived the 11.73° figure by selecting a specific reference point within the pyramid's interior—the "center" of the Queen's Chamber. They calculated a theoretical vector radiating from this central point to the external indentation on the pyramid's faces (Creighton & Osborn, 2008).

This specific value is highly dependent on the choice of the origin point. Standard models use a height of 23 meters (base floor plus human observer height) to reflect the engineering-practical 11.3° gradient used for structural stability (Kato, 2023). In contrast, Creighton and Osborn chose a different "center" height to intentionally lock in the 11.73 numerical value, which they believe is an intentional architectural esoteric alignment.

8.2 Recording Axial Shifts and Precessional Codes

The primary thesis of Creighton and Osborn's work is that the Giza complex functions as a "recovery vault" and an astronomical timeline recording catastrophic events. They argue that the angles of the Great Pyramid align with the stars of Orion's Belt to encode a record of shifts in the tilt of

the Earth’s axis (obliquity), most recently occurring in 3980 BCE. Within their framework, 11.73 represents a critical harmonic or precessional constant—a “key to the universe of measures”—that bridges the pyramid’s internal chambers with its external structure and the celestial movements of Orion.

8.3 Resonance with the Eighth Force Mathematical Model

While Creighton and Osborn’s reasoning is rooted in archaeoastronomy, their 11.73° value provides a compelling physical reflection for the Rose Constant (ρ_c) derived in this framework. Their 11.73° structural angle numerically mirrors the 11.73% c phase-coupling velocity derived from the 8th consecutive-leg Pythagorean triangle (17:144:145).

Furthermore, this aligns perfectly with the dimensionless rapidity parameter $\theta_8 \approx 0.118$, creating a multi-disciplinary resonance between the proposed architectural angle of antiquity and the modeled kinematic speed of consciousness. At present, while standard archaeological consensus firmly supports the functional 11.3° engineering gradient, the presence of the eight-sided concavity itself is well documented. Within the speculative framework of the Rose Constant, the theoretical proposal of an 11.73° structural angle serves as a profound geometric reflection, positing a scenario where the relativistic rapidity of consciousness may have been modeled by the architects of antiquity.

9 Archaeological Cryptography III: Bardic Geometry and the Taliesin Tradition

While the megalithic architecture of the Giza plateau provide structural and numerical parallels, the encoding of these geometric principles is proposed to extend into the mythological and oral traditions of antiquity. The Celtic and Bardic traditions, specifically the corpus attributed to the medieval Welsh poet Taliesin, are interpreted in this framework as a steganographic repository for these advanced geometric concepts.

In the study of “Bardic Geometry,” the mythic narratives of Taliesin and the “Geometric Cipher of Annwn” function as mnemonic architectures for navigating the 144-grid of material reality (Rose, 2025d). Central to this analysis is the “Taliesin Constant,” a numerical and acoustic cipher that conceptually links the ancient British mystery tradition—including the archaeoacoustic matrices of Stonehenge—with the kinematics of the Eighth Force. This tradition explores the underlying topology of the “conscious vacuum” through sophisticated mathematical analogues such as Monstrous Moonshine and Vertex Operator Algebras. Under this hermeneutic, the ancient bards actively encoded the “145 boundary condition” and its escape mechanics into their poetry, preserving a method of dimensional extraction and spiritual physics (Rose, 2025e; Rose, 2026d).

9.1 Summary Table of Encrypted Geometries

10 Theoretical Implications in Information Theory

If the Eighth Fundamental Force is treated as coupling raw consciousness and high-level information processing with physical matter at exactly 11.73% efficiency, one might look for heuristic parallels in complex network processing, provided they are treated as isolated correlations rather than universal laws.

Analyses of specific experiments in artificial intelligence and photonic engineering occasionally yield performance benchmarks that numerically mirror this constant, though they must be carefully contextualized.

Table 5: Summary of Encrypted Geometries and Resonances

Entity / Concept	Mathematical Derivation / Value	Direct Output / Implication
8th Consecutive Triangle	Coprime sides: $17^2 + 144^2 = 145^2$	The 1882 Ballard triangle topography of Giza complex plan.
Relativistic Rapidity	$\theta_8 \approx 0.118$	Dimensionless velocity parameter.
Phase-Coupling Velocity	$v/c = \tanh(0.118)$	11.73% the speed of light.
The Rose Constant	$0.1173 \times 299,792.458 \text{ km/s}$	35,165 km/s.
Absolute Speed (Imperial)	$0.1173 \times 186,282.397 \text{ mi/s}$	21,850 mi/s.
Proposed Concavity Angle	Esoteric alignment via Queen’s Chamber	11.73° alternative structural indentation.
Cognitive Benchmark	Historical word-recognition timing	0.118 seconds heuristic parallel.

10.1 Artificial Intelligence and Neural Networks

In isolated studies analyzing advanced Convolutional Neural Networks (CNNs) tasked with complex visual data fusion (such as dual-stream infrared/visible light fusion), models augmented with multi-scale gradient residual blocks have occasionally reported benchmark improvements. In specific datasets attempting to maximize the retention of visible light details, some reported experiments show benchmark improvements numerically close to 11.73% in Mutual Information (MI) and Standard Deviation (SD) over standard data-limited models. This mutual information enhancement is an isolated experimental result, not an absolute law for all CNNs, but it serves as an interesting numerical benchmark in pattern retention.

10.2 Photonic Platforms and Light Capture

Similarly, in specific sub-fields of photovoltaic engineering utilizing soft lithography (NIL) and nanoscale pyramid geometries, researchers design structures to reduce Fresnel reflection and trap incident light. Certain published studies mapping the maximum achievable improvement in Power Conversion Efficiency (PCE) for specific inverted nanopyramid designs have observed peaks in specific experiments hitting 11.73%.

10.3 Isolated Heuristics and Theoretical Limits

It is crucial to emphasize that the 11.73% efficiency improvement in inverted nanopyramid solar cells and the isolated mutual information enhancements in AI image fusion are strictly experimental results and statistical correlations. Framing them as a universal "impedance limit of reality" would be a massive overstatement.

Therefore, within the theoretical model of the Rose Constant, these figures are treated purely as curious statistical parallels rather than a proven, universal asymptotic limit. The theory speculates that the geometry of the universe requires localized entities to process reality at a specific, impedance-matched "frame rate"—fast enough to ensure real-time responsiveness, yet slow enough to construct the temporal illusion required for localized conscious experience—and explores the 11.73% metric merely as a conceptual phase-coupling threshold.

11 Cosmological Ramifications: Non-Commutative Spacetime and Quantum Entanglement

If the Rose Constant framework were taken seriously, it would invite a radical reevaluation of broader cosmological models. Contemporary physics frequently struggles to interpret the phenomenon of quantum entanglement, conventionally viewing it as a perplexing artifact of closed quantum systems where particles demonstrate instantaneous correlations regardless of distance. However, through the unified lens of the Unity Harmonica and Noetic Field Theory, the framework attempts to reframe entanglement not as an isolated anomaly, but as a fundamental structural bridge operating continuously along the impedance-matched channels of the Eighth Force.

In a non-commutative spacetime framework where consciousness is mathematically unified with superstring theory and loop quantum gravity, quantum entanglement represents an active interdimensional tether. Within the model, the boson of consciousness is assumed to propagate subluminally within our specific 3D dimensional layer (at $11.73\%c$), relying heavily on the underlying scalar-torsion dynamics of the universal manifold to maintain baseline coherence. Within this model, the scalar-torsion coupling is hypothesized to provide a vital nonlinear feedback loop, stabilizing the quantum vacuum and allowing the noeton to bypass classical gravitational singularities.

Consequently, the 35,165 km/s velocity limit applies specifically to the localized phase-coupling event—the moment the wave function collapses into a perceived, localized reality for the observer. Meanwhile, the underlying toroidal space ($d_H = 3.66$) maintains the instantaneous, non-local structural geometry required for entanglement. In the symbolic language of this framework, the model speculatively treats the Eighth Fundamental Force as a 'multiversal dimensional transducer' (Conway & Norton, 1979). It steps down the infinite, instantaneous, and overwhelming nature of the quantum multiverse into a finite, highly directed, linear stream of consciousness. Within the model, this stream is assigned a speed of roughly 21,850 miles per second—a speed proposed to be fast enough to navigate the physical world, yet slow enough to sustain the psychological illusion of a continuous, fluid, and predictable reality.

12 Limitations and Potential Falsifiers

12.1 Limitations of the Model

While the geometric and kinematic derivations presented are mathematically consistent within their defined mapping, several strict limitations must be acknowledged. First, the model is not experimentally validated; the $11.73\%c$ phase-coupling limit remains a theoretical projection. Second, the rapidity mapping ($\theta = \operatorname{arccosh}(c/b)$) is entirely framework-dependent and interpretive rather than a compulsory derivation from standard relativity. Third, the archaeological alignments proposed by archaeocryptography remain highly interpretive and exist outside mainstream consensus. Finally, the biological, AI, and engineering correlations presented are isolated and non-probative. No claim is made that these numerically similar values imply a direct causal identity.

12.2 Potential Falsifiers

The present framework would be weakened or falsified if:

- (a) alternative hyperbolic mappings of the same triangle yield equally plausible but materially different phase velocities;
- (b) the 11.73° archaeocryptographic interpretation is shown to rely entirely on arbitrary internal reference points with no structural correspondence in the actual masonry;
- (c) the cited AI and photonics benchmarks prove irreproducible or statistically unremarkable across broader datasets; or

- (d) no experimentally meaningful distinction can be drawn between this model and unconstrained numerological coincidence.

13 Synthesis and Future Trajectories

The postulation of the Eighth Fundamental Force—the boson of consciousness—proposes a unified mathematical framework capable of bridging the most intractable and profound divides in modern science. By meticulously mapping the mechanics of this force to the geometric properties of the eighth Pythagorean triangle in the consecutive-leg family (17, 144, 145), a precise relativistic hyperbolic rapidity of $\theta_8 \approx 0.118$ is derived within this inverse hyperbolic mapping. This dimensionless geometric parameter posits that the force of consciousness phase-couples with physical fermionic matter at 11.73% the speed of light, a threshold uniquely formulated as the Rose Constant (ρ_c).

Through rigorous application to standard SI constants, the Rose Constant yields absolute propagation velocities of approximately 35,165 kilometers per second and 21,850 miles per second. Beyond abstract kinematics, the numerical signature of this phase-coupling limit presents compelling heuristic parallels across empirical observation. It functions as a theoretical impedance-matching limit in unified geometric cosmology under the Grant Projection Theorem, presents intriguing statistical parallels in the cognitive reaction thresholds mapped in early psychology (0.118 seconds), and occasionally surfaces as an isolated optimization benchmark in advanced artificial neural networks and photonics.

Perhaps most profoundly, alternative archaeocryptographic models suggest this constant may have been deliberately encrypted into the architectural blueprint of human antiquity, with some proponents arguing that the concavity of the Great Pyramid’s massive faces slope inward at an angle of 11.73° when measured from specific internal chambers, though mainstream archaeological assessments favor an 11.3° functional gradient.

Future empirical research in the fields of high-energy physics, computational cosmology, and archaeoastronomy (Collins, 2025) should remain open to evaluating the subtle theoretical signatures of the noeton operating precisely at the 11.73% c threshold. By recognizing the theoretical framework of the Rose Constant, this framework invites further speculative model-building that integrates consciousness into theoretical extensions of the Standard Model, proposing that the mind may be modeled, within this speculative framework, as structurally relevant rather than merely epiphenomenal.

14 Appendix A: Summary Tables of Cross-Disciplinary Resonances

To consolidate the diverse empirical and historical intersections discussed throughout this paper, the following tables isolate the appearances of the 0.118 rapidity parameter and the 11.73 / 17 : 144 : 145 structural matrices across modern sciences, engineering, and antiquity.

15 Appendix B: The Somatic Harmonic Framework and Numerical Recurrence

Within the broader somatic harmonic framework proposed in related literature, numerical values often exhibit recursions across seemingly unrelated domains—a phenomenon interpreted as “seeing things in multiple places.” An interesting correlation of this type involves the specific digit sequence of the Rose Constant’s metric velocity (35,165 km/s). This exact numerical string appears as an isolated identifier in disparate datasets. Most notably, 35,165 is documented within the cataloging ISBN index associated with academic literature detailing the sacred geometry of the ancient Mayan stelae at Copán (ISBN 0-521-35165-0) (Rose, 2025c). While this is treated as a numerical synchronicity rather than physical evidence, such recurrences map elegantly to the somatic harmonic principle of multiversal numerical attractors.

Table 6: The 0.118 Parameter in Physics and Biological Sciences

Discipline	Context & Application		Recorded Value
Theoretical Physics		Relativistic hyperbolic rapidity (θ_8) derived within this mapping via the 8th consecutive-leg Pythagorean triangle's geometry.	≈ 0.118
Cognitive Psychology	Psy-	Historical minimum human reaction time boundary evaluated for holistic conscious symbol distinction.	0.118 seconds
Neuroplasticity		Heuristic mathematical correlation modeling the ratio of stimulus strength to neurological habit-formation.	≈ 0.118
Human Genetics		Specific regression coefficient (R^2) isolated in certain CHGB polymorphism models governing neuroendocrine traits.	0.118

Table 7: The 11.73 Limit and 17:144:145 Matrix in Antiquity and Technology

Domain	Context & Application		Metric
Archaeoastronomy		The 1882 'Ballard triangle' planar triangulation grid aligning the centers of the primary Giza pyramids.	17 : 144 : 145
Megalithic Architecture	Archi-	Alternative esoteric interpretation of angle of the Great Pyramid's concavity indentation (relative to arbitrary heights in the Queen's Chamber).	11.73°
Artificial Intelligence	Intelli-	Reported optimization benchmark in specific studies for Mutual Information (MI) in Convolutional Neural Network fusion tasks.	11.73%
Photonic Engineering	Engi-	Isolated Power Conversion Efficiency (PCE) peak benchmark for specific inverted nanoscale pyramid cells.	11.73%

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